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학 력	성균관대학교 전자전기컴퓨터공학과 박사학위 취득예정 (2027.02)		
Google Scholar	scholar.google.com/citations?user=GhwtxtgAAAAJ		
병 역	전문연구요원 복무만료 (2017.03 – 2020.03)		
주 소			

전문분야 (Research Coverage) — 로봇 파운데이션 모델(RFM) 등 모델과 플릿 스케줄러 등 알고리즘이 환경이 변해도 QoE를 최대화하도록 모델 입력·모델 자체를 제어하고, 조건이 변하면 운영 에이전트가 도구·스케줄러를 선택하는 시스템을 개발함. 강화학습 정책 전이와 LLM 에이전트로 구현함. 전용 시뮬레이터·성능 예측 모델·테스트베드를 구축해 검증하고 시스템 5건을 전달함

① Reinforcement Learning — Resource Management & Remote Model Serving

: 강화학습으로 모델 입력의 전송률·대역 할당 등 자원 관리 정책을 학습하고, 애플리케이션의 최종 성능(QoE) 기준이나 네트워크 환경이 변화해도 재학습 없이 또는 소량 데이터로 전이하는 알고리즘을 개발해 시뮬레이터·실기기 테스트베드에서 검증

- **XferQoE**(2024–2026, IEEE IoT Journal 심사 중, 제1저자) — 모델 입력의 전송률을 정하는 단일 제어기를 애플리케이션 정보 없이 네트워크 품질 지표(QoS)만으로 사전학습하고, 재학습 없이 QoS 항목별 비중만 맞춰 애플리케이션별 QoE를 최대화함
  - ✧ 정책 추론은 엣지 서버가 수행하고 관측만 실제 링크를 건너는 원격 모델 서빙 구조에서, NS-3와 실기기 테스트베드에 사전학습 제어기를 각각 전이하여 QoE를 평가함 (사용 로봇 시뮬레이터: Meta-World·ManiSkill3)
- **Repot**(2020–2021, IEEE Access 2021, 제1저자 · 제1발명자 특허 한·미 등록) — 센싱 디바이스들이 클라우드 서버에 업데이트하는 정보의 품질을 최대로 유지하기 위해, 네트워크 환경 변화에 적응하도록 대역폭 할당 제어를 소규모 데이터로 재학습함
- **원격 로봇 파운데이션 모델(RFM) 서빙**(XferQoE 후속 연구) — 이기종 협업-피지컬AI 과제(2026–2030)와 연계, 클라우드의 RFM이 실기기를 구동할 때 요청 지연이 성공률과 처리량에 미치는 영향을 지연 주입 평가 환경(LIBERO · MIKASA)으로 측정함

② Agentic AI — LLM Agents for System Operation (Robotics & V2G)

: 조건 변화를 감지해 도구·스케줄러를 선택하고 요구조건을 코드로 변환해 검증하는 LLM 에이전트 설계·구축(V2G 운영 · 로봇 조작)

- **LDuS**(2023–2024, NeurIPS 2024, 제2저자) — 학습된 로봇 skill diffusion 정책이 학습 때 없던 조건·요구조건을 재학습 없이 따르게 하는 zero-shot 정책 적응 프레임워크로, 정책 전이와 LLM 자동화 두 축을 결합함
  - ✧ 언어로 주어진 미학습 제약(속도·에너지·공간 등)을 LLM이 손실 함수 코드로 변환해 로봇 궤적 생성을 유도하고, 생성된 코드와 궤적을 LLM이 피드백 루프로 검증해 잘못된 코드는 재생성하고 궤적이 어긋나면 유도 적용 빈도를 높임
- **AgentV2G**(2026.03–2027.03) — 현대자동차·기아 V2X 과제 실무책임자로 설계한 전기차 플릿의 전력시장 입찰·충방전 운영 프레임워크로, 시뮬레이터 V2G-OpSim(제주 이중 정산 시장 규칙)을 함께 개발함
  - ✧ 전력 가격과 차량 이용 두 축의 조건 변화를 LLM 에이전트가 감지해 가격 축은 가격 전망을 보정하는 도구를, 차량 축은 조건별로 학습한 강화학습 스케줄러를 선택하고, 적합한 도구가 없으면 새 도구를 생성함. 과제 중간보고에서는 1,000대로 학습한 차량 공통 정책을 1만 대에 재학습 없이 적용해 주기당 66ms에 충·방전을 계산함

③ Systems Engineering — Simulators, Testbeds, Protocols & Deployment

: 학습 모델을 배포 환경에 맞게 설계·학습하고, 모델 학습·검증용 시뮬레이터·테스트베드와 프로토콜·서비스를 구축함

- 학습 모델 구축 — 엣지 NPU 객체 인식 모델: 주간 표지판 데이터로 연합학습(FedAvg)한 YOLOv8n에 야간 표지판 데이터로 사전학습한 YOLOv8l을 지식증류(backbone P5 증류 층)해 야간 정확도를 연합학습만 대비 최대 20%p 높이고, FPGA NPU 실시간 추론과, 확신도가 낮으면 다음 프레임을 서버의 YOLOv8l로 보내는 협업 추론으로 구동 · NVMe-oF 성능 예측 모델: FIO 실측으로 학습한 DNN이 노드별 IOPS·지연을 동시에 예측하며 장치 구성 4종에서 실측과 정합
- 시스템 구축 — 네트워크 시뮬레이터·테스트베드: 패킷 수준 C++ 시뮬레이터에서 학습한 제어 정책을 단일 인터페이스로 NS-3·802.11ac 16노드 테스트베드에 그대로 옮겨 검증 · 삼성전자 전달 2건: 4K 60 Hz 무선 스트리밍용 TCP·신뢰성 UDP 송수신 시스템, 시스템 사양만으로 재구성되는 NVMe-oF 스토리지 시뮬레이터(미국 특허 2건) · IoT 프로토콜: 정의·명세, 고정 크기 자원 식별자 패킷 재설계(삼성전자 공동 특허) · PS-LTE 망 선택: 실측 20만 건 기반 대역 선택 API·앱(Polaris AI 전달)

기술 역량

Languages — C/C++, Python, MATLAB
Machine Learning — JAX/Flax, PyTorch, reinforcement learning (meta-RL, offline RL, in-context adaptation, weight-conditioned multi-objective control, successor features), domain adaptation, representation learning (variational sequence encoder–decoder, graph embedding, Transformer encoders), time-series/world-model prediction, diffusion policies, knowledge distillation, federated learning (Flower, FedAvg)
LLM & Agentic Systems — Code-as-Policies, critic-based closed-loop verification
Simulation & Environments — custom packet-level network simulators, vehicular mobility simulation (SUMO), V2G fleet-operation environments, DNN surrogate modeling of measured systems (NVMe-oF storage, FIO benchmarking)
Remote Model Serving & Resource Management — networked robot control (Meta-World · ManiSkill3), off-board policy inference, robot-task QoE modeling, RFM fine-tuning, RFM serving under network delay, VLA fine-tuning (π0.5, SmolVLA), MAE-pretrained ViT encoders

Networking — TCP/UDP protocol implementation (NS-3), IEEE 802.11 family, LTE / PS-LTE, qdisc queue modeling
Systems & Deployment — multithreaded C/C++, Raspberry Pi deployment, YOLO detection on FPGA-based edge NPUs

학력사항 및 경력사항

기간	학교 및 전공	비고
2013.03 – 2027.02 (학위취득예정)	성균관대학교 대학원 전자전기컴퓨터공학과 석박사통합과정 박사학위논문: Reinforcement Learning for Quality-Driven Network Control 2018 우홍욱 교수 연구실 이동 · 학위논문 주제 확정	GPA 4.33 / 4.5 · 박사과정 수료 2017.02 지도교수: 우홍욱 · CSI-Agent Group
2017.03 – 2020.03	성균관대학교, Software Convergence Research Institute	전문연구요원, 3년 의무복무 만료
2008.03 – 2013.02	성균관대학교 전자전기공학부 (공학사)	GPA 4.23 / 4.5

주요 산학 · 정부 연구과제

기간	과제명	발주기관	역할
2026.03 – 2027.03	V2X 운영 최적화를 위한 강화학습 기반 기술 개발	현대자동차 · 기아	실무책임자
2025.09 – 2025.12	피지컬AI 핵심기술 PoC — 이기종 협업-피지컬AI SDF 특화 기반 모델 연구개발	과기정통부 · NIPA	참여연구원
2024.08 – 2024.12	재난망·철도망 전파간섭 지역 AI 기반 PS-LTE 단말 접속 개선 연구	Polaris AI	실무책임자
2021.04 – 2025.06	데이터센터-엣지 NPU 연합 추론/학습 SW 프레임워크 개발	과기정통부 · IITP	실무책임자
2019.08 – 2022.07	차세대 컴퓨팅환경을 위한 스토리지시스템 및 응용프레임워크 최적화 기술 연구	삼성전자	실무책임자
2017.09 – 2018.09	저 지연 무선 네트워크 프로토콜 개발	삼성전자	실무책임자

이 외 국방과학연구소 · 과학기술정보통신부(구 미래창조과학부) · 교육부 국가 R&D 과제 14건 수행 (연구책임자 1 · 실무책임자 5 · 참여연구원 8)

연구실적 요약

시스템 개발 4개 기관에 5건 전달 — 삼성전자(×2) · Polaris AI · Acryl · 국방과학연구소, 전 건 실무책임자

논문(2018년 이후) SCIE 국제저널 3편(제1저자 2) · NeurIPS 2024 Main Track(제2저자) · IEEE IoT Journal 심사 중(제1저자) (2012–2018 국내외 논문 35편: B급 국제저널 3 · A급 국내저널 5 · B급 국제학술대회 18 · 국내학술대회 9, Google Scholar 참조)

특허 등록 8건(국내 3 · 미국 5), 발명 6건 — 제1발명자 발명 2건(한·미 각 등록) · 삼성전자 공동 등록 3건

대표 논문

[심사 중] Youngseok Lee and Honguk Woo, "XferQoE: QoS-to-QoE transfer for adaptive rate control," submitted to IEEE Internet of Things Journal.

[NeurIPS] Woo Kyung Kim, **Youngseok Lee**, Jooyoung Kim, and Honguk Woo, "LLM-based skill diffusion for zero-shot policy adaptation," Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Main Conference Track.

[SCIE] Youngseok Lee, Woo Kyung Kim, Sung Hyun Choi, Ikjun Yeom, and Honguk Woo, "Repot: Transferable reinforcement learning for quality-centric networked monitoring in various environments," IEEE Access, vol. 9, 2021.

[SCIE] Minjae Park, **Youngseok Lee**, Ikjun Yeom, and Honguk Woo, "Gemma: Reinforcement learning-based graph embedding and mapping for virtual network applications," IEEE Access, vol. 9, 2021.

[SCIE] Youngseok Lee, Suin Lee, Chan-Gun Lee, Ikjun Yeom, and Honguk Woo, "Continual prediction of bug-fix time using deep learning-based activity stream embedding," IEEE Access, vol. 8, 2020. — 인용 33회

등록 특허 (8건)

등록번호	발명의 명칭	비고
KR 10-2629731 US 12,047,250 B2	Graph Embedding-Based Virtual Network Mapping Method	제1발명자, 한·미 등록
KR 10-2574070 US 12,556,492 B2	Network Monitoring System	제1발명자, 한·미 등록
US 12,210,760 B2	Object Storage System, Migration Control Device, and Migration Control Method	삼성전자 공동(발명자 4/4)
US 12,650,769 B2	Simulation Method and Modeling Method	삼성전자 공동(발명자 4/4)
US 10,742,742 B2	Electronic Apparatus, System for Internet-of-Things Environment and Control Method Thereof	삼성전자 공동(발명자 3/4)
KR 10-1305307	Simulation Apparatus and Method for Time Synchronization in Spread Spectrum System	GNSS 연구(발명자 3/3)

장학 및 수상

기간 / 연도	내역	수여기관
2013.03 – 2018.02	Global Ph.D. Fellowship — 전 기간(5년) 수혜, 2013년 전국 200명 선발	교육부 · 한국연구재단
2008.03 – 2013.02	국가우수장학금(이공계) — 8학기 전액	교육과학기술부 · 한국과학재단
2014, 2015	Outstanding Research Achievement, Global Ph.D. Fellowship Program	교육부 · 한국연구재단
2012.12	IEEE Seoul Section Student Paper Contest — Bronze Award	IEEE Seoul Section

성균관대학교

RESEARCH PORTFOLIO

QoE-Driven Control of Models and Algorithms under Environment Shift

Two control targets for QoE (quality of experience): model input (transmission rate, frame quality) and model adaptation; under drift an operating agent selects the tool or scheduler

① **Reinforcement Learning** — Resource Management & Remote Model Serving
② **Agentic AI** — LLM Agents for System Operation (Robotics & V2G)
③ **Systems Engineering** — Simulators, Testbeds, Protocols & Deployment

Youngseok Lee

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Youngseok Lee — Ph.D. Candidate, Sungkyunkwan Univ.

- **Education** — Integrated M.S.–Ph.D., Electrical & Computer Engineering, Sungkyunkwan University (Mar. 2013 – Feb. 2027, expected) · Advisor: Prof. Honguk Woo (CSI-Agent Group) · Dissertation: Reinforcement Learning for Quality-Driven Network Control
  - Global Ph.D. Fellowship, 5-year term (2013–2018, Ministry of Education / NRF Korea) · Expert Research Personnel service (Mar. 2017 – Mar. 2020) · 2018: moved to Prof. Honguk Woo's group and defined the dissertation topic · 35 earlier publications (2012–2018, two other groups: Prof. Ikjun Yeom's and Prof. Seokho Yoon's)
- **Achievements** — 5 systems delivered to Samsung Electronics ×2, Polaris AI, Acryl, and ADD (Agency for Defense Development), each as technical leader · since 2018: NeurIPS 2024 (second author), 3 SCIE journal papers (first author 2 / second author 1), IEEE IoT Journal (first author, under review) · 8 granted patents (US 5 / KR 3; six inventions, first inventor on two inventions, KR and US each) · 20 funded projects (principal investigator 1 · technical leader 10 · researcher 9)
  - (full cover) in a Role line marks an item whose every listed role the applicant carried

Research area	Representative work	Outcomes
① **Reinforcement Learning** — Resource Management & Remote Model Serving	○ XferQoE — A QoS-pretrained controller of a model's input data rate, adapted per application by objective-weight fitting from QoE feedback ○ XferQoE with robot foundation models (π0.5, SmolVLA; NS-3 Wi-Fi, page 7) ○ Physical-AI PoC — Code-as-Policies demo · robot foundation model (RFM) fine-tuning on a real arm (SO-101) ○ Repot — Bandwidth allocation transferred across network environments by flow embedding and action shaping	○ IEEE IoT Journal (first author, under review) ○ IEEE Access 2021 (first author) ○ 1 first-inventor patent family (KR 2023 · US 2026)
② **Agentic AI** — LLM Agents for System Operation (Robotics & V2G)	○ LDuS — LLM-guided skill diffusion for zero-shot policy adaptation ○ AgentV2G — An LLM agent that detects price and vehicle-use drift and selects the matching tool (price-forecast correction, condition-specific RL schedulers) for V2G (vehicle-to-grid) fleet bidding and dispatch; SAC EV dispatch policy	○ NeurIPS 2024 (second author)
③ **Systems Engineering** — Simulators, Testbeds, Protocols & Deployment	○ S-Sim packet simulator → NS-3, 16-node Raspberry Pi Wi-Fi testbed ○ Code-as-Policies demo pipeline · SO-101 real-arm fine-tuning setup ○ V2G fleet-operation environment (market rules · SMP · EV sessions) ○ FL with knowledge distillation → edge-NPU YOLOv8n inference pipeline ○ NVMe-oF storage simulator with a DNN surrogate of IOPS and latency ○ LTE band-selection service · QLED wireless streaming testbed ○ IoT protocol specification; fixed-size identifier packets (patent)	○ Systems delivered ×5 (Samsung ×2, Polaris AI, Acryl, ADD) ○ 3 US patents with Samsung (storage ×2 · IoT resource identification)

Delivered Systems and Testbeds — Recipient, Period, Role

Delivered systems (5) and built systems (3)

System	Delivered to	Period	Role	Page
Network Environments of Increasing Fidelity for Rate-Control Policy Pretraining and Validation (S-Sim, NS-3, NiL Testbed)	—	2020–2026	Leader	7
V2G Fleet-Operation Evaluation Environment for Two-Settlement Markets (V2G-OpSim)	—	2026	Leader	10
Edge–Server Collaborative Inference Pipeline for Real-Time Traffic Sign Recognition	Acryl	Apr. 2021 – Jun. 2025	Leader	13
NVMe-over-TCP Disaggregated Storage Simulator with a Dual-Output DNN Surrogate	Samsung Electronics	Aug. 2019 – Jul. 2022	Leader	14
Database-Backed Network Selection Application for Mobile Terminals	Polaris AI	Aug. – Dec. 2024	Leader	15
Low-Latency Wireless Streaming Testbed for the Samsung QLED TV One Connect System	Samsung Electronics	Sep. 2017 – Sep. 2018	Leader	15
GUI-Based Analysis Tool for Direct GPS P(Y) Code Acquisition under Wideband Jamming	ADD	Mar. – Oct. 2013	Leader	CV
Unified JAX/Flax Training Framework for Network Control Policies	—	2020–2026	Leader	CV

ADD = Agency for Defense Development · NiL = network-in-the-loop · S-Sim = Simplified Simulator · CV = described in the CV only

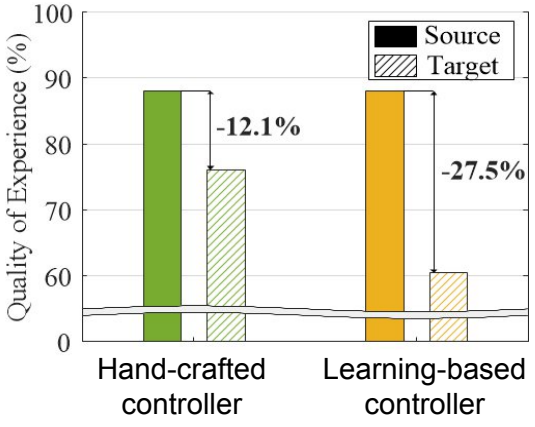
Contents — Domain, Target, and Techniques by Page

Work	Domain	Target	Techniques	Page
Research Coverage	Overview of the shared problem (QoE-driven control under environment shift) and the three research axes			5
XferQoE	Remote model serving	Image-frame quality for a remote model, through network rate control	Meta-RL, Transformer	6
XferQoE	Remote model serving	Robot observations sent to an off-board policy or RFM over Wi-Fi	Sim2Real, RFM	7
Physical AI	Robot manipulation	RFM fine-tuning on a real arm, and task success under serving delay	RFM, VLA, VLM	8
LDuS	Robot manipulation	Zero-shot adaptation of a skill policy to unseen language-specified contexts	Diffusion, Action Decoder	9
AgentV2G	V2G fleet operation	Charging-hub market bids and EV dispatch under price and vehicle-use drift	RL Scheduler, LLM	10
Repot	Networked monitoring	Bandwidth allocation among sensing devices sharing one link	RL, Fine-Tuning	11
XferQoE	Networked monitoring	Camera sending rates on one shared link to a server's recognition model	Object Recognition, Real-World Testbed	12
Federated Edge Inference	Edge AI	Federated training and edge–server inference of a traffic-sign recognition model	Federated Learning, Knowledge Distillation	13
NVMe-oF Storage Simulator with a DNN Surrogate	Data-center storage	IOPS and latency of disaggregated NVMe-oF storage, reproduced in software	Supervised Learning, Regression	14
PS-LTE Network Selection	Public-safety networks	Which network a terminal accesses, PS-LTE or a commercial LTE band	PostGIS	15
QLED Streaming Testbed	Wireless video streaming	Latency of 4K 60 Hz video over the QLED TV's wireless link	TCP/UDP	15
Protocol Definition for IoT Devices	Internet of Things	Packet bytes spent identifying IoT devices	MQTT, Packet Design	16
Publications, Patents, and Demo Videos	Selected papers and patents with links, demo videos, and contact			17

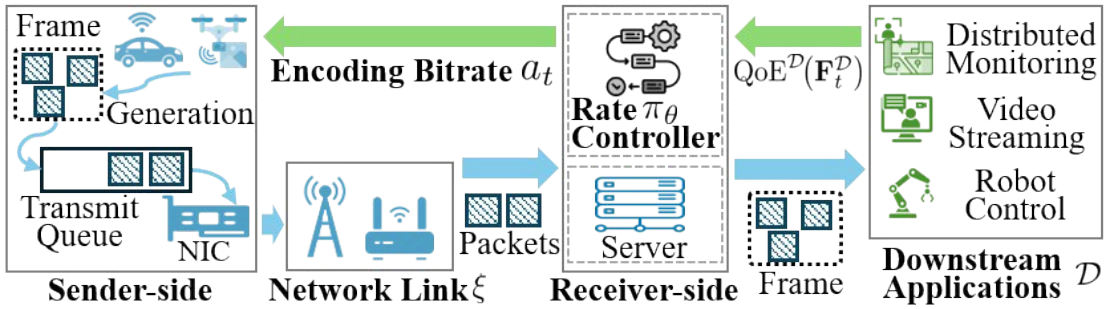
RFM = robot foundation model · VLA = vision-language-action model · VLM = vision-language model · EV = electric vehicle · PS-LTE = public-safety LTE

QoE-Driven Control of Models and Algorithms under Environment Shift

- **Problem** — A learned controller matches a hand-crafted one under the conditions it was trained for. Yet when those conditions shift even slightly after deployment, it loses far more QoE (quality of experience), while the hand-crafted one degrades gracefully (left figure)
  - Such shifts occur in the application's QoE definition or in the network condition, electricity price, or vehicle-use pattern. For each condition, the hand-crafted controller needs manual re-tuning. The learned one needs data collection and retraining
- **Goal** — Maximize the QoE of models such as robot foundation models (RFMs) and of algorithms such as fleet schedulers under environment shift. Two control targets serve this goal: model input (transmission rate, frame quality) and model adaptation. Under drift, an operating agent also selects the tool or scheduler. Learned controllers should combine hand-crafted robustness with learned performance and adapt from a small amount of data without retraining the deployed policy
  1. **Reinforcement Learning** — A meta-RL rate controller of a model's input and a bandwidth-allocation policy for devices sharing one link, both transferring to new objectives and environments with little or no retraining, and an RL dispatch policy for an electric-vehicle fleet that is independent of the fleet size
  2. **Agentic AI** — An LLM agent that detects drift in a system's operation and selects or creates the tool or scheduler for the new condition. An LLM also writes loss-function code from a user's requirement to steer a pretrained robot policy without retraining
  3. **Systems Engineering** — Purpose-built simulators, surrogate models, and testbeds on which the two approaches are implemented and validated. Five systems built with the same engineering were delivered to four organizations
- **Representative system** — QoE-driven control loop (right figure): RL-based rate controller → network link → application QoE feedback



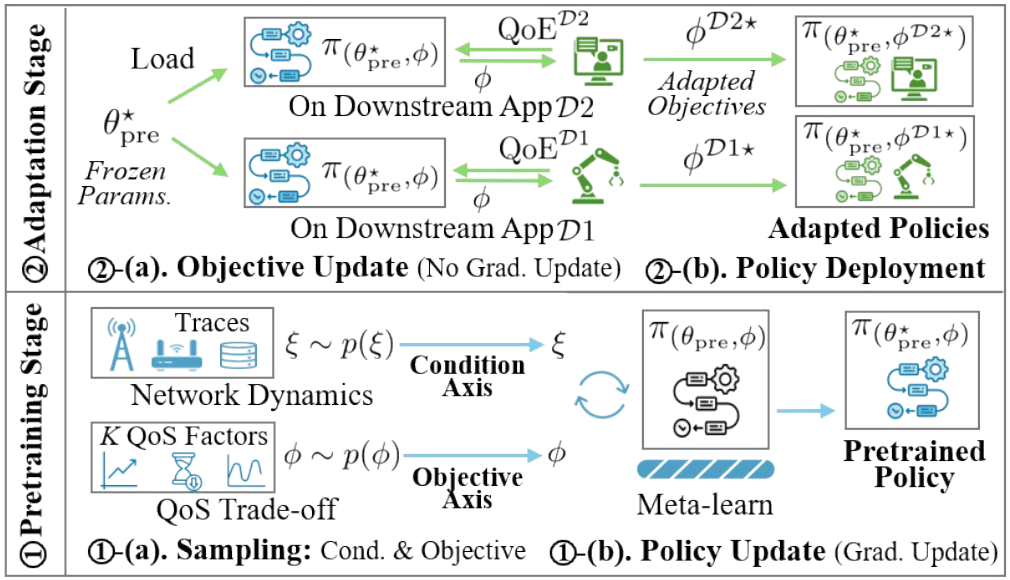
Motivating measurement (Repot)



Closed-loop QoE-driven adaptive rate control (XferQoE)

[Meta-RL, Transformer] XferQoE — QoS-Pretrained Rate Control for Model Serving

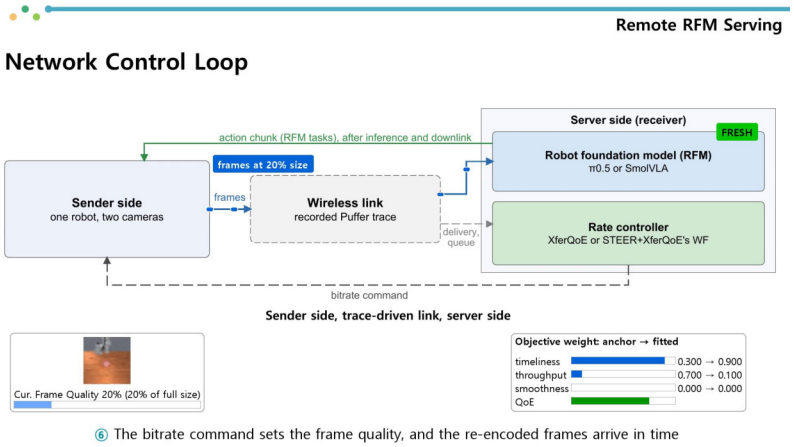
- **Achievements** — IEEE IoT Journal (first author, under review)
- **Role: Leader** — (full cover) problem definition, algorithm design, implementation, experiments, testbed
- **Target system** — Remote model serving. A robot or a camera streams observation frames over a wireless link to a server-side model, which receives only the frames the link can carry
- **Problem** — A controller trained per application repeats both the objective design and the full-stack QoE (quality of experience) collection
- **Approach** — A pretrained rate controller sets the sending rate of the served model's input, and with it the quality. The model itself stays untouched
  - **Algorithm** — Meta-RL (meta-reinforcement learning). Stage 1: the controller is meta-pretrained on QoS (quality of service) objectives alone in S-Sim (Simplified Simulator). Each episode samples the two meta-training axes, a bandwidth condition and an objective weight w, so the policy learns to infer the condition from recent history. Stage 2: the policy is frozen. Only w is fitted from a few adaptation episodes of QoE feedback on each application, with no gradient update (figure)
  - **Structure** — SAC (soft actor-critic) with one module per axis. A regime–context encoder (a bidirectional Transformer and a per-patch 1-D CNN) infers the condition as two latents. Factor critics (one Q-head per QoS factor) compose the value linearly under w. The actor reads the state, both latents, and w



QoS pretraining → QoE adaptation

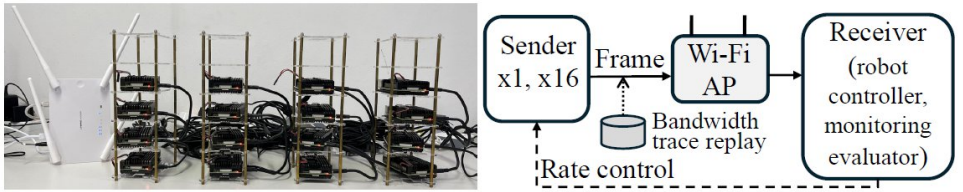
[Sim2Real, RFM] XferQoE — Real-Link Testbed and Off-Board RFM Serving

- **Target system** — The edge server runs the manipulation policy and the rate controller, and only observations cross the real wireless link. The robot runs in simulation (ManiSkill3, Meta-World), so the Sim2Real target is the network side
- **Approach** — Design: three network environments of increasing fidelity share one state-input, rate-output interface and replay the same bandwidth trace, packet-level C++ S-Sim (Simplified Simulator) → full-stack NS-3, the NiL (network-in-the-loop) testbed. The same frozen rate controller runs on all three with its parameters unchanged. On NS-3 Wi-Fi, the policy can also be a robot foundation model (RFM), a vision-language-action (VLA) model that computes each action chunk from the newest frame arriving within its deadline. The same controller then sets the RFM's input rate and refits only its objective weights per task (lower-right figure)
- **Implementation** — The NiL testbed: one Raspberry Pi 4 sender for robot control (up to 16 for monitoring) on a real 802.11ac link via an access point to the edge server. C/C++ daemons implement episode-level clock synchronization and frame-deadline expiry accounting (upper-right figure)

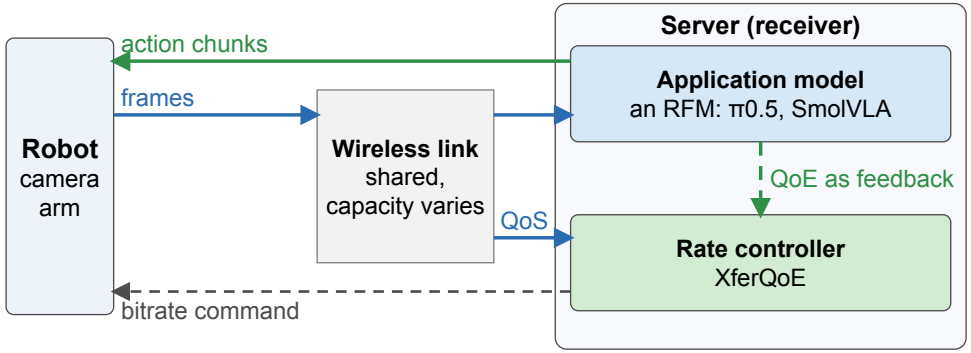


Remote RFM serving: control loop and two RFM tasks

sonny0714.github.io/demo_video_robot/xferqoe-rfm-serving.html



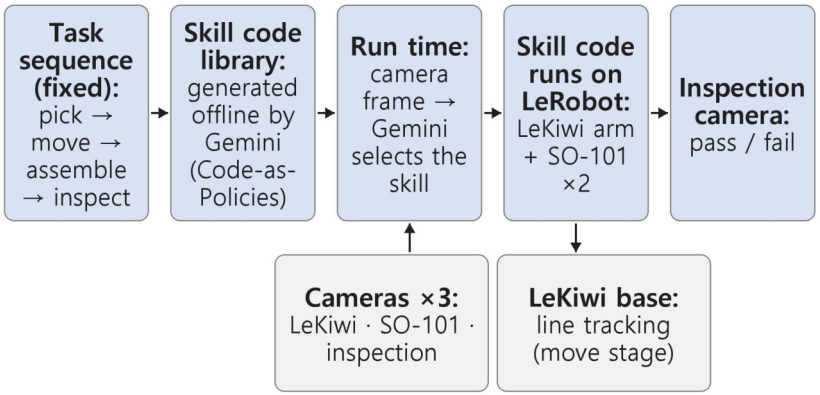
NiL testbed: Raspberry Pi 4 senders, Wi-Fi AP, edge server



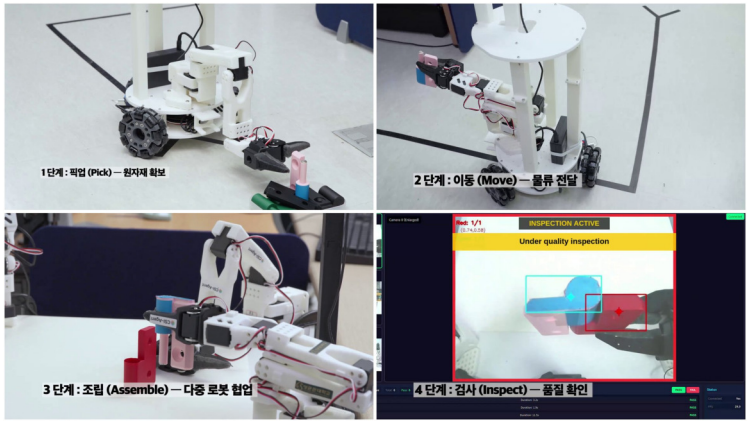
Off-board RFM serving with rate control at the server

[RFM, VLA, VLM] Physical AI — RFM Fine-Tuning, Serving under Delay, PoC Demo

- **Robot foundation model (RFM) encoder fine-tuning on a real arm (SO-101)**
  - **Role: Leader** — implementation, experiments (encoder pretraining, data collection, fine-tuning, real-arm tests)
  - **Target system** — H2R (human-to-robot) transfer on the arm: human videos guide imitation learning, so only the vision encoder is trained
  - **Approach** — Design: H2R was reproduced on public data, where ViT (Vision Transformer) encoders pretrained by masked autoencoding served as the backbone of ACT (Action Chunking with Transformers). Then the vision encoders of SmolVLA and π0.5 were fine-tuned with the action expert frozen
  - **Implementation** — Donut pick-and-place demonstrations collected by leader-arm teleoperation. Fine-tuning and real-arm trials on LeRobot
- **RFM serving under network delay (simulated; delay-injection harness)**
  - **Role: Leader** — problem definition, experiments (research direction, experiment design, and analysis)
  - **Target system** — An off-board RFM serving a robot: a static task (LIBERO, π0.5) and a moving-ball task (MIKASA, SmolVLA)
  - **Approach** — Design: fixed or variable delay is injected at the chunk request, where uplink, inference, and downlink delays add up, so one value covers the serving path. Each policy runs asynchronously (switching chunks) or synchronously (waiting for the next chunk)
- **Physical-AI PoC demo (real robots, Code-as-Policies)**
  - **Role: Participant** — pipeline structure design, LeRobot runs
  - **Target system** — A four-stage demo (pick → move → assemble → inspect) on two SO-101 arms and a LeKiwi mobile base
  - **Approach** — Design: the task sequence is fixed. Gemini, a VLM (vision-language model), generates each skill's code offline for a skill library. At run time it selects the skill from the camera frame, and the selected code drives the arms on LeRobot (left figure, video)



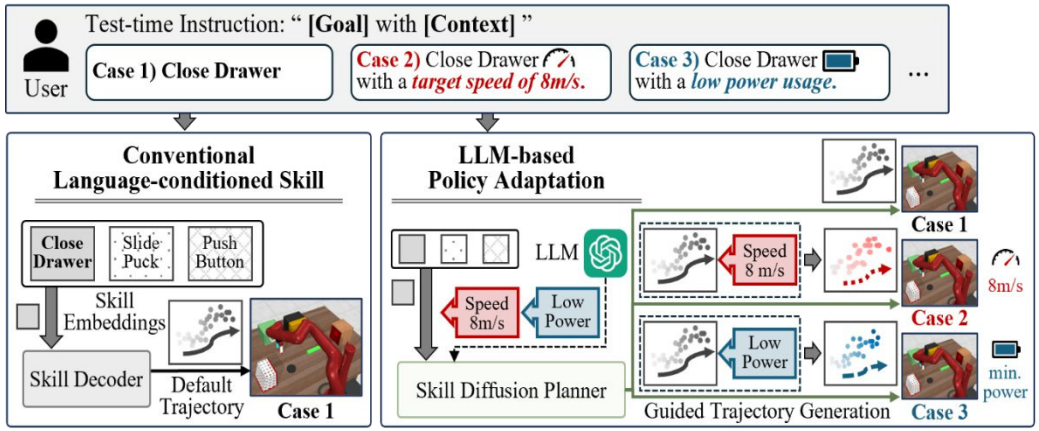
Demo: offline skill code; camera → Gemini selects skill → LeRobot



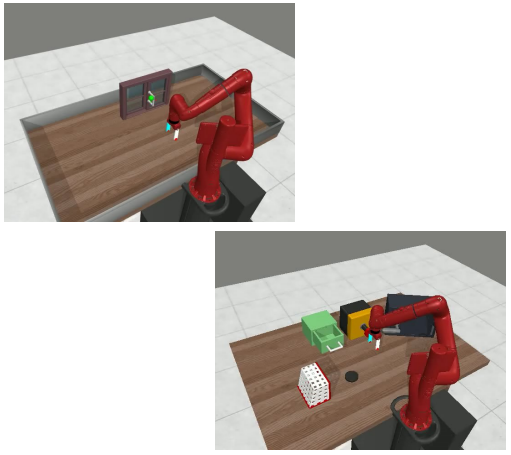
PoC demo video (41 s): pick, move, assemble, inspect

[Diffusion, Action Decoder] LDuS — LLM-Guided Skill Diffusion for Policy Adaptation

- **Achievements** — NeurIPS 2024 Main Conference Track (second author)
- **Role: Leader** — problem definition, algorithm design, architecture, experiment design; robot-task validation on Meta-World
- **Target system** — Manipulation policies that follow a natural-language context, such as speed, power usage, or a spatial constraint
- **Problem** — A skill-based policy cannot adapt to a context unseen in training. Retraining per context is expensive (left figure, Cases 2–3)
- **Approach** — Unlike Code-as-Policies, where an LLM writes executable policy code, here the LLM writes a loss function. Loss-guidance code generated from the context steers the pretrained policy at test time, with no policy update
  - **Algorithm** — Pretraining uses goal-annotated expert trajectories without context. In adaptation, GPT-3.5 writes loss-guidance code from the context, and the gradient of that loss guides every denoising step. GPT-4 judges contact from the camera frame to gate constraints that apply only while touching the object. A closed loop validates the code and the guided trajectory. It regenerates faulty code and raises the guidance frequency until the context is satisfied
  - **Structure** — A VAE skill encoder maps an h-step state–action segment to a skill embedding. A skill prior predicts the embedding from the state and the language goal encoded by CLIP (contrastive language–image pretraining). A skill diffusion planner (a U-Net denoiser as in Diffuser) reconstructs the segment. Sequential in-painting fixes the past states and actions into the generated trajectory



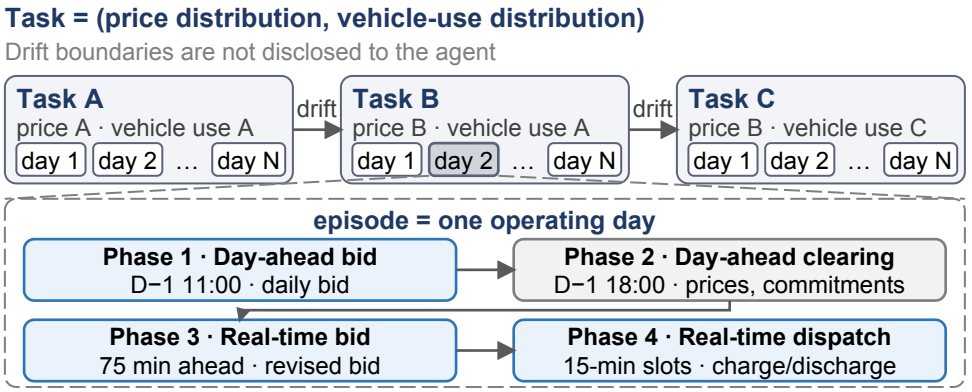
LDuS: language context → LLM-generated loss code → guided trajectory generation



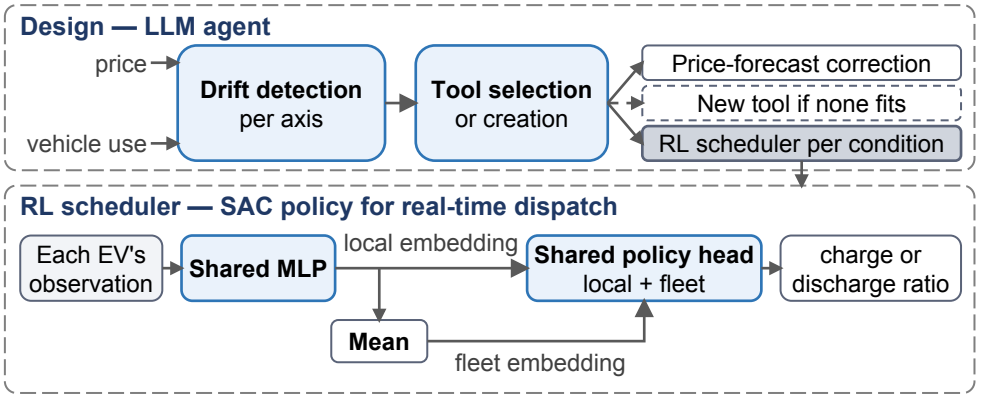
Manipulation task executions (Meta-World)

[RL Scheduler, LLM] AgentV2G — Agentic V2G Fleet Scheduling Under Drift

- **Achievements** — Hyundai Motor · Kia V2X (vehicle-to-everything) project · interim report delivered Oct. 2026
- **Role: Leader** — (full cover) problem definition, algorithm design, implementation, experiments; construction of the V2G fleet-operation environment
- **Target system** — A V2G (vehicle-to-grid) charging hub. It bids in a day-ahead and real-time two-settlement electricity market and dispatches the charging and discharging of its electric vehicles (EVs)
- **Problem** — An agent free to re-plan cannot be trusted to commit irreversible bids. A fixed pipeline cannot follow the drift (left figure)
- **Approach** — An LLM agent makes a fixed bidding-and-dispatch pipeline follow the drift
  - **Design** — The agent detects drift on two axes, electricity price and vehicle use, when observations that differ from the reference accumulate past a set amount. On the price axis it selects a tool that corrects the price forecast. On the vehicle axis it selects an RL scheduler trained for the new condition. When no fitting tool exists, it creates one from the accumulated observations (right figure)
  - **Structure** — The condition-specific scheduler is a SAC (soft actor-critic) policy for the real-time dispatch stage. A shared MLP embeds each EV's observation, and the mean of these embeddings forms the fleet embedding. A shared policy head reads both and emits each EV's charge or discharge ratio, so the policy is independent of the fleet size. The action is learned as a residual over rule-based dispatch and can be projected onto the allowed band
- **Implementation** — V2G-OpSim, the charging-hub simulator built on the Jeju two-settlement pilot rules, with EU and NEM (Australian National Electricity Market) presets. KPX (Korea Power Exchange) SMP (system marginal price) series and EV charging-session records drive it



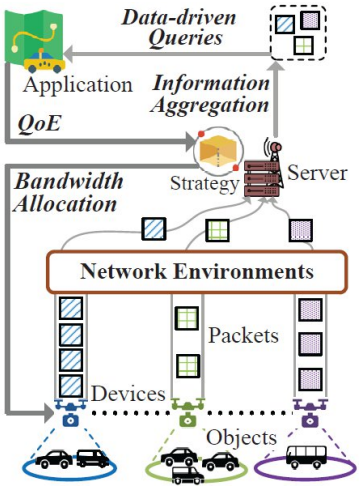
Task stream with two-axis drift and one operating day's phases



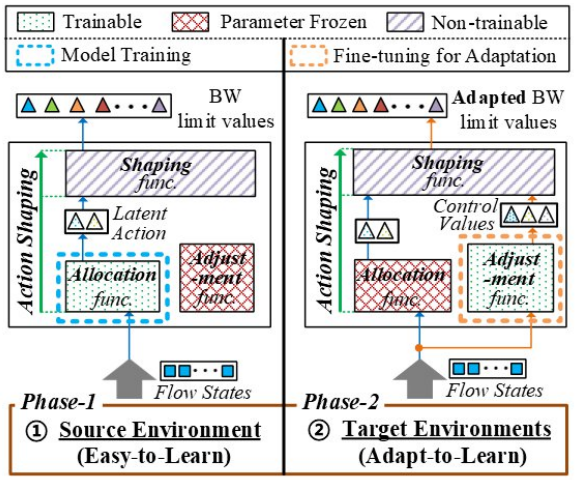
Drift detection, tool selection, and the RL scheduler

[RL, Fine-Tuning] Repot — Transferable RL for Networked Monitoring

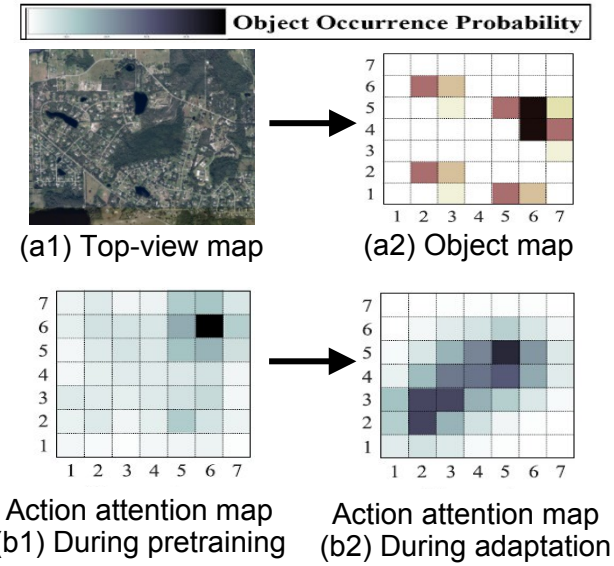
- **Achievements** — IEEE Access 2021 (first author) · Network Monitoring System patent, KR 10-2574070 · US 12,556,492 B2 (first inventor)
- **Role: Leader** — problem definition, algorithm design, implementation, experiments
- **Target system** — Networked monitoring. When sensing devices share a single link to a server, the bandwidth allocation must keep the server's continual query accurate as the network degrades (left figure)
- **Problem** — Re-learning that allocation from scratch for each new network scale and condition is prohibitively expensive
- **Approach** — Train where samples are inexpensive and adapt where they are expensive, by updating a small adjustment function alone
  - **Algorithm** — Two-phase transfer (center figure). Phase 1: the policy is trained end to end with SAC (soft actor-critic) in the time-slotted source environment. Phase 2: in the NS-3 target, all learned modules are frozen, and a second, small SAC policy (the adjustment function with its own critics) is trained. Its outputs enter the shaping function together with the frozen latent action
  - **Structure** — Flow embedding: a Transformer encoder (per-device MLP vectorization + self-attention over all flows) represents every stream by its history and its relation to the others. Allocation: a trainable function emits a latent action. Shaping: a non-trainable function maps it to per-device bandwidth limits by a distance-weighted softmax (right figure: action attention maps)



Networked monitoring systems



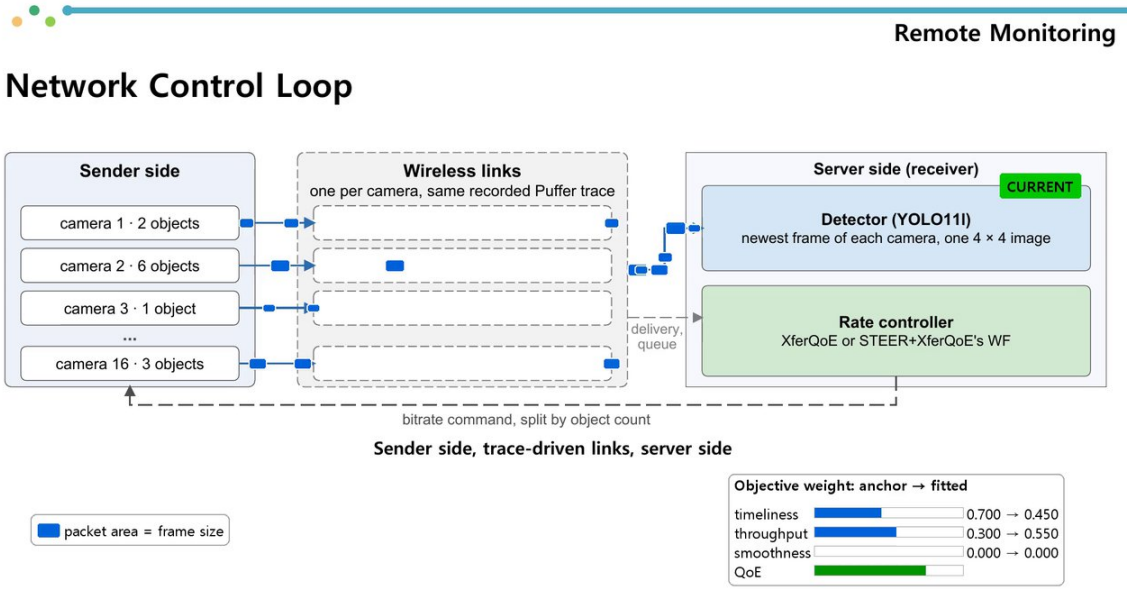
Action shaping model structure



Action attention map transfer

[Object Recognition, Real-World Testbed] XferQoE — Networked Monitoring

- **Target system** — Networked monitoring. Many cameras share one wireless link to a server whose recognition model reads their frames. A late frame shows a past scene, and a heavily compressed frame loses small objects
- **Approach** — Design: the same frozen rate controller sets the cameras' sending rates. Only its objective weights are refitted to the recognition QoE (quality of experience). Repot (page 11) targets this kind of system through policy fine-tuning instead
- **Implementation** — Sixteen Raspberry Pi 4 senders on the real-world testbed of page 7 act as sixteen cameras. They stream pedestrian-scene frames over one 802.11ac link to the edge server, which runs a recognition model (YOLO11) and scores every interval by detection F1 against the current ground truth. The same scoring runs on NS-3 Wi-Fi. S-Sim (Simplified Simulator) uses a recall model fitted to the recognition model



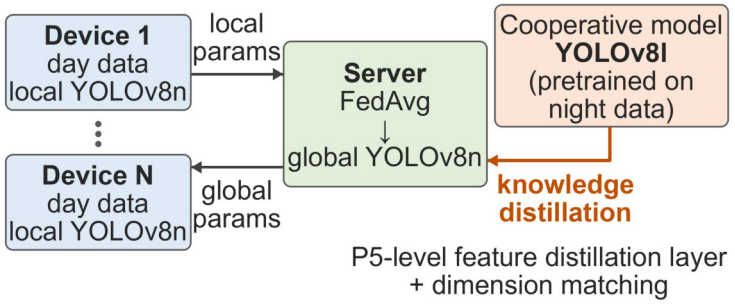
⑥ The bitrate command raises each camera's frame quality, split by object count, and the frames still arrive in time

Networked monitoring: control loop, then one trace run (145–185 s)

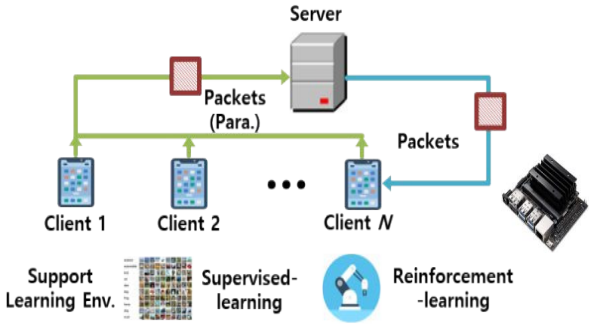
sonny0714.github.io/demo_video_robot/xferqoe-monitoring.html

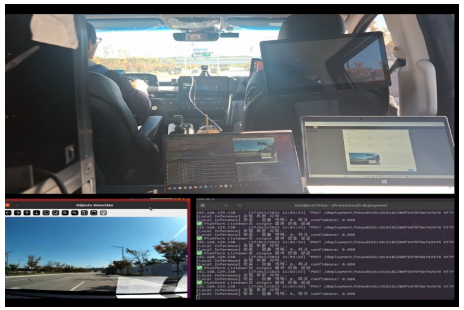
[Federated Learning, Knowledge Distillation] Federated Edge Inference

- **Achievements** — Delivered to Acryl Inc.
- **Role: Leader** — problem definition, algorithm design, implementation, experiments, testbed
- **Target system** — Edge AI: real-time traffic-sign recognition on edge NPUs (neural processing units) backed by a data-center server
- **Problem** — A global model federated from the devices' daytime data generalizes poorly to night, and night data is assumed scarce on the devices
- **Approach**
  - **Design** — Cooperative federated learning with knowledge distillation (left figure). After each FedAvg (federated averaging) aggregation, the server distills a cooperative large model (a YOLOv8l pretrained on night traffic-sign data) into the global YOLOv8n, without moving the devices' data. The NPU model handles routine frames. After a low-confidence frame, the next frame goes to the server's YOLOv8l
  - **Structure** — A distillation layer on the backbone's P5-level features with a dimension-matching projection
- **Implementation** — An FL (federated learning) testbed on the open-source Flower framework (right figure (a)). Its client side was extended with federated RL and per-client non-IID (non-independent, non-identically distributed) data composition. A per-round learning-rate schedule and image augmentation were added. The YOLO models and the inference system were deployed on Acryl's Jonathan server and an FPGA-based NPU (FPGA: field-programmable gate array). The partners field-tested them in a vehicle (right figure (b))



Cooperative Federated Learning with YOLO Models

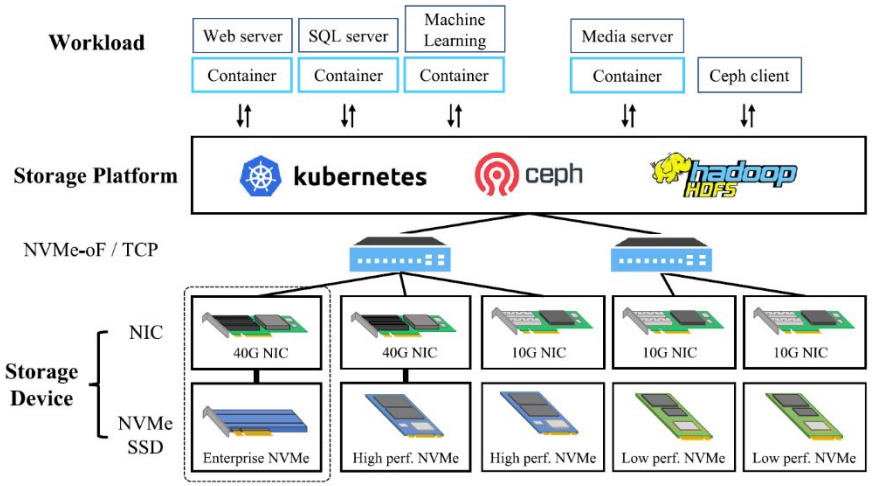




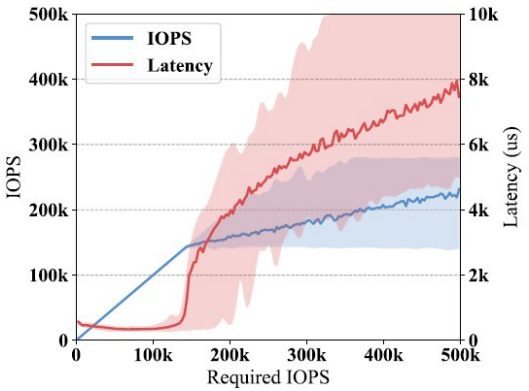
NPU project: (a) embedded FL testbed · (b) in-vehicle field test

[Supervised Learning, Regression] NVMe-oF Storage Simulator with a DNN Surrogate

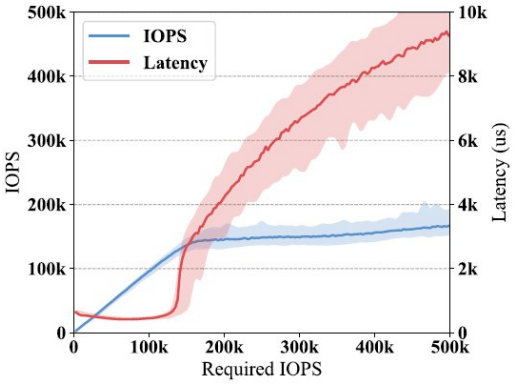
- **Achievements** — Delivered to Samsung Electronics · US 12,650,769 B2 · US 12,210,760 B2 (fourth of four inventors on both)
- **Role: Leader** — problem definition, algorithm design, experiments
- **Target system** — Data-center storage: a disaggregated NVMe-oF (NVMe over Fabrics; here NVMe/TCP) cluster, reproduced in software (based on NS-3) so that storage-management policies (RL policies included) can be tuned for mixed configurations of heterogeneous devices
- **Approach**
  - **Design** — FIO (flexible I/O tester) measures per-node serviced IOPS (I/O operations per second) and latency against required IOPS. One DNN (deep neural network) surrogate spanning device specifications reproduces both responses (right figure: (a) measured, (b) estimated)
  - **Structure** — The surrogate takes a histogram of the objects' required IOPS and predicts serviced IOPS and latency with one DNN. Devices are organized under a storage platform that aggregates them in a two-level hierarchy, so the environment is reconfigured from a system specification rather than rebuilt (left figure)
- **Implementation** — Four device configurations were profiled with FIO, from an RC500 SSD on 10 GbE to a Samsung PM1725a SSD on 40 GbE. A tier-level object-migration policy was evaluated across homogeneous and heterogeneous compositions under FIO workloads



NVMe-oF disaggregated storage systems



(a) Measured performance pattern

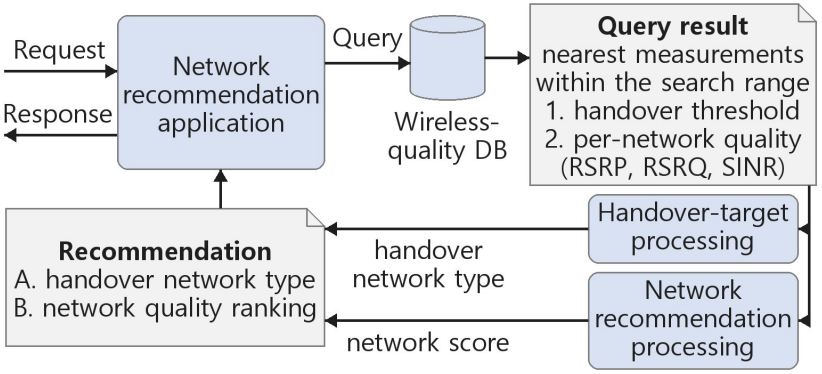


(b) Estimated performance pattern

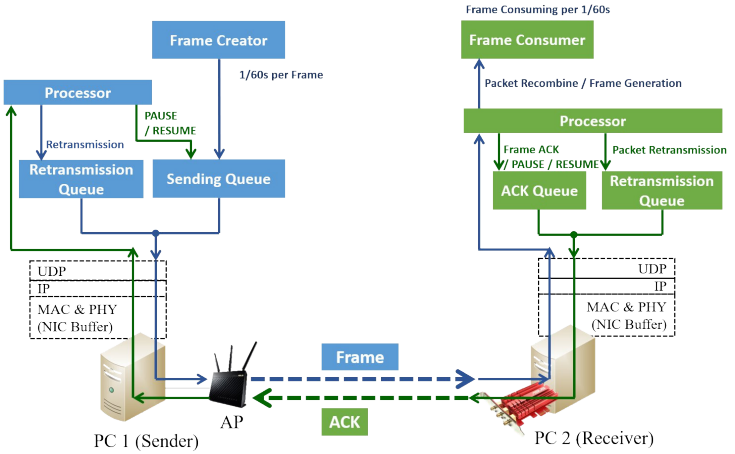
A surrogate model example for NVMe-oF storage

[PostGIS, TCP/UDP] PS-LTE Network Selection · QLED Streaming Testbed — Delivered Systems

- **Database-backed network selection application for PS-LTE (public-safety LTE) terminals**
  - **Achievements** — Delivered to Polaris AI
  - **Role: Leader** — algorithm design, implementation
  - **Target system** — A terminal choosing PS-LTE or a commercial band where the disaster-response and railway networks interfere
  - **Approach** — Design: the application ranks the LTE bands by signal strength and quality (RSRP, RSRQ, SINR) from a database of normalized field measurements. The PS-LTE–commercial handover is decided by thresholds
  - **Implementation** — A PostGIS database, a Flask server with an HTTP API, and a Flutter web application on the terminal (left figure)
- **Low-latency wireless streaming testbed for the Samsung QLED One Connect system**
  - **Achievements** — Delivered to Samsung Electronics
  - **Role: Leader** — implementation, experiments, testbed
  - **Target system** — The wireless link that feeds 4K 60 Hz video from the One Connect box to a QLED TV
  - **Approach** — Design: layer-by-layer delay analysis of the standard TCP/UDP stack, with bitrate-paced UDP against TCP flow control
  - **Implementation** — Transmitter and receiver Linux hosts with BCM4360 802.11ac NICs. C/C++ components: a TCP sender–receiver, a UDP sender–receiver, and a raw-socket sniffer for RTT (round-trip time), throughput, loss, duplication, and reordering (right figure)



PS-LTE network selection application



One Connect wireless streaming testbed

[MQTT, Packet Design] Protocol Definition for IoT Devices

- **Achievements** — US 10,742,742 B2 with Samsung Electronics (third of four inventors)
- **Role: Leader** — algorithm design (protocol specification and packet-structure redesign)
- **Target system** — Internet of Things (IoT): the message protocol between an intelligent gateway and its devices
- **Problem** — Each packet carries the device's long, variable-length URI-style identifier in full and wastes network capacity (left figure)
- **Approach** — Design: a controlling apparatus assigns one fixed-size identifier item per URI level. The device stores it, requests and commands carry it, and both ends map it in a resource access table (left figure). Packets then carry the fixed-size identifier instead of the URI, so they shrink. The gateway forwards each device its own access ID (a boiler and a CCTV camera in the right figure)
- **Implementation** — The gateway–device protocol was defined as an MQTT-based message specification with the redesigned packet structure

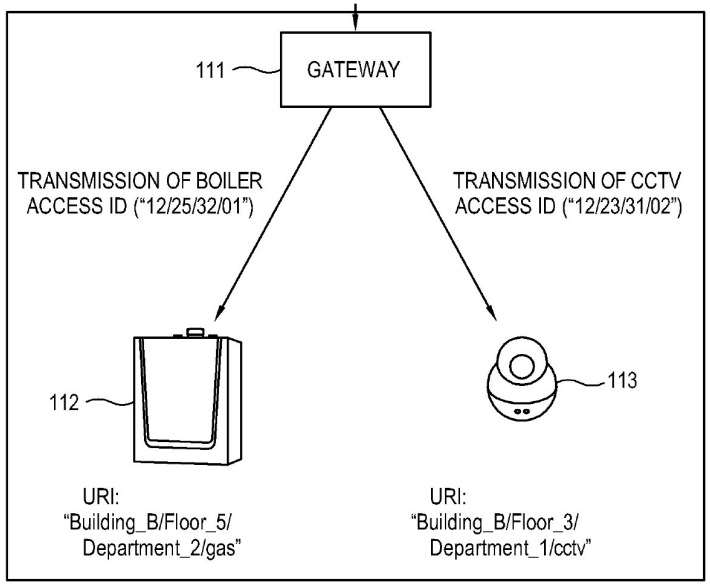
URI	/Building_A/Floor_B/Department_C/Office_D/temperature_4
RESOURCE ACCESS ID	/25/61/78/81/04

(a) URI AND RESOURCE ACCESS ID

RESOURCE ACCESS ID	URI
04	temperature_4
25	Building_A
61	Floor_B
78	Department_C
81	Office_D

(b) RESOURCE ACCESS TABLE

Fixed-size access IDs for URIs



Gateway forwarding access IDs

Publications, Patents, and Demo Videos — Contact

- **Contact** — Youngseok Lee · yslee.gs@gmail.com · +82-10-7368-6845 · scholar.google.com/citations?user=GhwtxtgAAAAJ

Selected publications and patents (links)

Type	Title	Venue · Registration	Link
Paper · under review	XferQoE: QoS-to-QoE transfer for adaptive rate control	IEEE IoT Journal · first author	—
Paper · top-tier	LLM-based skill diffusion for zero-shot policy adaptation (LDuS)	NeurIPS 2024 Main Conference Track · second author	NeurIPS
Paper · SCIE	Repot: Transferable reinforcement learning for quality-centric networked monitoring in various environments	IEEE Access, vol. 9, 2021 · first author	DOI
Paper · SCIE	Gemma: Reinforcement learning-based graph embedding and mapping for virtual network applications	IEEE Access, vol. 9, 2021 · second author	DOI
Paper · SCIE	Continual prediction of bug-fix time using deep learning-based activity stream embedding (DASENet)	IEEE Access, vol. 8, 2020 · first author 33 citations	DOI
Patent · first inventor	Graph Embedding-Based Virtual Network Mapping Method	KR 10-2629731 (2024) US 12,047,250 B2 (2024)	KR · US
Patent · first inventor	Network Monitoring System	KR 10-2574070 (2023) US 12,556,492 B2 (2026)	KR · US
Patent · with Samsung	Object Storage System, Migration Control Device, and Migration Control Method	US 12,210,760 B2 (2025)	US
Patent · with Samsung	Simulation Method and Modeling Method	US 12,650,769 B2 (2026)	USPTO
Patent · with Samsung	Electronic Apparatus, System for Internet-of-Things Environment and Control Method Thereof	US 10,742,742 B2 (2020)	US

Demo videos (public pages at sonny0714.github.io/demo_video_robot/)

Demo	What the video shows	Link
PokeCube: STEER+XferQoE's WF vs. XferQoE (5× speed)	Played in turn over the same 41.7 s: STEER+XferQoE's WF finishes two tasks, XferQoE five	xferqoe-pokecube.html
Remote RFM serving: control loop, LIBERO-Spatial (π0.5), MIKASA Intercept (SmolVLA)	The control loop on the system diagram, then both RFM tasks, first under STEER+XferQoE's WF and then under XferQoE	xferqoe-rfm-serving.html
Networked monitoring: control loop, one trace run	The control loop on the system diagram, then one trace run (145–185 s): at 148.5 s camera 7 finds four of its five objects under XferQoE and none under STEER+XferQoE's WF	xferqoe-monitoring.html

STEER+XferQoE's WF = STEER with XferQoE's objective-weight fit

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RESEARCH SUMMARY

- **Achievements**
  - **5 systems delivered** to 4 organizations (Samsung Electronics ×2, Polaris AI, Acryl, and the Agency for Defense Development), each as technical leader
  - **Since 2018** (Prof. Honguk Woo's group, CSI-Agent Group) — dissertation *Reinforcement Learning for Quality-Driven Network Control*; NeurIPS 2024 (second author), IEEE Access 2020 and 2021 (first author ×2) and 2021 (second author), and a first-author IEEE IoT Journal submission under review
  - **8 granted patents** (US 5 / KR 3) covering six inventions — first inventor on two inventions (KR and US each), three US patents (storage ×2, IoT) held jointly with Samsung Electronics
- **Path** — Global Ph.D. Fellowship, 5-year term (2013–2018) · Expert Research Personnel service (Mar. 2017 – Mar. 2020) · 2018: moved to Prof. Honguk Woo's group and defined the dissertation topic · 35 earlier publications (2012–2018, two other groups: Prof. Ikjun Yeom's and Prof. Seokho Yoon's) on Google Scholar
- **Research Coverage** — The work builds control systems that maximize the QoE (quality of experience) of models such as robot foundation models (RFMs) and algorithms such as fleet schedulers under environment shift, through two control targets, model input (transmission rate, frame quality) and model adaptation, and through the tool or scheduler an operating agent selects under drift. RL policy transfer and LLM agents provide that control; both are implemented and validated on purpose-built simulators, surrogates, and testbeds, and five systems were delivered (Systems Engineering).

Research Map

Research area	Representative work	Outcomes
① Reinforcement Learning — Resource Management & Remote Model Serving	○ XferQoE — A QoS-pretrained meta-RL controller of a model's input data rate, adapted per application by objective-weight fitting ○ Repot — Bandwidth allocation transferred across network environments by flow embedding and action shaping ○ RFM (robot foundation model) encoder fine-tuning and real-arm evaluation (π0.5 and SmolVLA on SO-101) ○ Off-board RFMs under network delay — delay-injection evaluation harness (XferQoE follow-up) ○ Skill-based Code-as-Policies demo for a Physical-AI PoC	○ IEEE IoT Journal (first author, under review) ○ IEEE Access 2021 (first author) ○ 1 first-inventor patent family (KR·US)
② Agentic AI — LLM Agents for System Operation (Robotics & V2G)	○ LDuS — LLM-guided skill diffusion for zero-shot policy adaptation ○ AgentV2G — An LLM agent that detects price and vehicle-use drift and selects the matching tool (price-forecast correction, condition-specific RL schedulers) for V2G fleet bidding and dispatch	○ NeurIPS 2024 (second author)
③ Systems Engineering — Simulators, Testbeds, Protocols & Deployment	○ Packet-level C++ training simulator (S-Sim) → NS-3, 16-node Raspberry Pi Wi-Fi (NiL) testbed with an edge server ○ Code-as-Policies demo pipeline and SO-101 real-arm RFM fine-tuning setup ○ V2G fleet-operation environment — market rules, system marginal price (SMP) series, and EV charging-session records ○ Federated learning with knowledge distillation → YOLOv8n edge-NPU traffic-sign inference with a server-side YOLOv8l ○ NVMe-over-TCP storage simulator with dual-output DNN surrogate ○ LTE band-selection service · QLED wireless streaming testbed ○ IoT network protocol — specification; fixed-size identifier packets	○ Delivered ×5 (Samsung ×2, Polaris AI, Acryl, ADD) ○ 3 US patents with Samsung (storage ×2 · IoT identifier packets)
Deep Learning Models (2019–2021)	○ Gemma — Data-center and ISP virtual network mapping learned end to end in one framework ○ DASENet — Bug-fix time re-estimated continually from developer activity streams	○ IEEE Access 2021 (second author) · 2020 (first author) ○ 1 first-inventor patent family (KR·US)

TECHNICAL SKILLS

Languages	C/C++, Python, MATLAB
Machine Learning	JAX/Flax (JIT compilation, custom training loops), PyTorch, reinforcement learning (RL), meta-RL, offline RL, in-context adaptation, domain adaptation, weight-conditioned multi-objective control, successor features, representation learning, variational sequence encoder-decoder, graph embedding, time-series prediction, world-model prediction, diffusion policies, Transformer encoders (self-/cross-attention), knowledge distillation, federated learning (Flower, FedAvg)
LLM & Agentic Systems	Code-as-Policies, critic-based closed-loop verification
Simulation & Environments	custom packet-level network simulators, vehicular mobility simulation (SUMO), V2G fleet-operation environments, DNN surrogate modeling of measured systems (NVMe-oF storage, FIO benchmarking)
Remote Model Serving & Resource Management	networked robot control (Meta-World · ManiSkill3), off-board policy inference, robot-task QoE modeling, RFM fine-tuning, RFM serving under network delay, VLA fine-tuning (π0.5, SmolVLA), MAE-pretrained ViT (Vision Transformer) encoders
Networking	TCP/UDP protocol implementation (NS-3), IEEE 802.11 family, LTE / PS-LTE, qdisc queue modeling
Systems & Deployment	multithreaded C/C++, NPU/FPGA inference, embedded deployment (Raspberry Pi), YOLO-based object detection on edge NPUs (FPGA-based), edge–server collaborative inference

EDUCATION & MILITARY SERVICE

Period	Degree & Institution	Notes
Mar. 2013 – Feb. 2027 (expected)	Integrated M.S.–Ph.D., Electrical & Computer Engineering, Sungkyunkwan University — Dissertation: *Reinforcement Learning for Quality-Driven Network Control*	GPA 4.33 / 4.5 · Ph.D. coursework completed Feb. 2017 · Advisor: Prof. Honguk Woo, CSI-Agent Group (csiagentgroup.com)
Mar. 2017 – Mar. 2020	Military Service — Expert Research Personnel (전문연구요원), Software Convergence Research Institute, Sungkyunkwan University	Full 3-year term completed
Mar. 2008 – Feb. 2013	B.S., Electronic & Electrical Engineering, Sungkyunkwan University	GPA 4.23 / 4.5

DEVELOPMENT EXPERIENCE

Nineteen items across the three research axes and the 2019–2021 deep-learning model work — systems, models, protocols, and evaluation environments. In a Role line, (full cover) marks an item whose every listed role the applicant carried.

Reinforcement Learning — Resource Management & Remote Model Serving

[Meta-RL, Transformer, SAC] QoS-to-QoE Transfer Framework for Adaptive Rate Control (XferQoE)
2024 – 2026

- **Achievements** — IEEE IoT Journal, **first author** (under review)
- **Role: Leader** — (full cover) problem definition, algorithm design, implementation, experiments, testbed.
- **Built** — A rate controller sets the input data rate of each application's model over the network: pretrained on QoS (quality-of-service) factors alone, it is then adapted by fitting only the application's objective weights.
- **Model** — SAC (soft actor-critic) with one module per meta-training axis: a regime–context encoder (a bidirectional Transformer with mean pooling → regime latent; a per-patch 1-D CNN with a cross-attention readout → step latent, 8-dim each) and factor critics (one Q-head per QoS factor, ensemble of 3) that compose the value linearly under the objective weight w; the actor reads the state, both latents, and w (policy ≈ 0.3 MB, encoder ≤ 3.5 MB). Meta-pretraining samples a task per episode over the bandwidth dynamics and w in S-Sim. A predictive decoder (next state, QoS vector, change direction, future-window statistics) trains the encoder on the network dynamics alone; in an encoder-only probe on eight bandwidth traces the encoder leads PatchTST, TimesNet, and TimeMixer in next-step direction accuracy (78.0% against 75.1% for Repot's Transformer, the next best, on real-world traces).
- **Design** — Robot control, video streaming, and networked monitoring score a delivered stream differently, and their quality feedback is costly to collect. Pretraining needs none of it, and adaptation uses a few feedback episodes without updating the policy, because value is composed per QoS factor under the application's weighting.
- **Result** — From a single pretraining run, adapted QoE (quality of experience) in the simulator averages 91% of the QoE of the per-application end-to-end Oracle (a controller trained on that application's own QoE), and 95–102% on the robot-manipulation tasks. The adapted controller improves QoE over the strongest learning-based baseline within six adaptation episodes, and the transfer holds from the training simulator to NS-3 and a real-device testbed.

With robot foundation models as the policy on NS-3 Wi-Fi (π0.5 on LIBERO-Spatial, SmolVLA on MIKASA Intercept), it retains 0.88 and 0.51 of the zero-delay success, against at most 0.68 and 0.41 for the other controllers.

[Transformer, SAC, Policy Transfer] Transferable RL System for Quality-Centric Networked Monitoring (Repot) 2020 – 2021

- **Achievements** — IEEE Access (2021), **first author** · KR 10-2574070 · US 12,556,492 B2 (**first inventor**)
- **Role: Leader** — problem definition, algorithm design, implementation, experiments.
- **Built** — Repot allocates bandwidth among sensing devices sharing one link, so a server's continual query stays accurate as the network degrades.
- **Model** — Flow embedding by a Transformer encoder (per-device MLP vectorization + self-attention over all flows) → a trainable allocation function emitting a latent action → a non-trainable shaping function mapping it to per-device bandwidth limits by a distance-weighted softmax. The policy is trained end to end with SAC in the time-slotted source environment, then frozen while a second, small SAC policy (the adjustment function, with its own critics) is trained in the NS-3 target; its outputs enter the shaping function with the frozen latent action.
- **Design** — Each device's stream is represented by its own history and its relation to the others, so the policy transfers across network scales and conditions by updating a small adjustment function alone. Training takes place in an ideal time-slotted environment where samples are inexpensive; the policy then adapts to an NS-3 environment where that experience would be prohibitive to collect, and its QoE holds as the device count grows.
- **Result** — The policy adapts to a target network environment with 6.25% of the samples required for training from scratch.

[π0.5, SmolVLA, ACT, MAE-Pretrained ViT] Robot Foundation Model Encoder Fine-Tuning and Real-Arm Evaluation (SO-101) 2026

- **Role: Leader** — implementation, experiments (encoder pretraining, data collection, fine-tuning, real-arm tests).
- **Built** — Donut pick-and-place data from leader-arm teleoperation on a low-cost 6-DoF arm: 104 episodes (45,553 frames). On LeRobot (PyTorch), the vision encoders of the pretrained SmolVLA and π0.5 were fine-tuned with the action expert frozen; ACT (Action Chunking with Transformers), trained on the same episodes, was the baseline.
- **Model** — ACT: CNN vision backbone + transformer encoder–decoder trained as a conditional VAE; SmolVLA: SmolVLM-2 vision-language backbone + flow-matching action expert; π0.5: PaliGemma-based VLM + flow-matching action expert. Training ran 100K, 50K, and 30K steps, respectively (batch 8, 8, and 1).
- **Design** — The work targets human-to-robot (H2R) transfer, in which human videos guide robot imitation learning, so the vision encoder is the part trained. H2R was first reproduced on public data (two ViT-B/16 encoders pretrained by masked autoencoding, one on human videos and one on manipulation videos, each serving as the ACT backbone and compared offline by L1/MSE against an ImageNet-pretrained encoder), and the real-arm stage applied the same approach to the two pretrained RFMs (SmolVLA and π0.5).
- **Result** — On the real arm, 10 donut pick-and-place trials each at trained (in-distribution) and new (out-of-distribution) positions: ACT 70% / 40%, SmolVLA 90% / 50%, π0.5 80% / 60%.

[VLA (π0.5, SmolVLA), Action Chunking] Delay-Injection Evaluation Harness for Off-Board Robot Foundation Models 2026 – present

- **Role: Leader** — problem definition, experiments (research direction, experiment design, and analysis).
- **Built** — The harness injects fixed or per-request variable delay into chunk requests on LIBERO (static tabletop, π0.5) and MIKASA (moving ball, fine-tuned SmolVLA) and runs each policy asynchronously (switching to the new chunk) or synchronously (waiting for it) at delays of up to 1,000 ms.
- **Design** — Delay is injected at the chunk request, where the uplink, the inference budget, and the action downlink of the RFM add up, so one delay value covers the serving path; both execution modes run without delay compensation, and a moving-ball task is added because a static scene may hide the cost of delay.
- **Result** — Without delay compensation, delay costs success or time: at 500 ms, asynchronous execution falls from 97% to 80% on the static task while synchronous execution keeps its success rate but loses 35% of throughput (10.95 → 7.09 successes per minute); on the moving-ball task both fall and synchronous execution drops from 55.5% to 1.0%; variable delay does not differ significantly from fixed delay of equal mean.

[VLM (Gemini), LeRobot] Skill-Based Code-as-Policies Demo for a Physical-AI PoC (SO-101 Arms and a LeKiwi Base) 2025 – 2026

- **Role: Participant** — pipeline structure design, LeRobot runs.
- **Built** — A four-stage demo (pick → move → assemble → inspect) on two SO-101 arms and a LeKiwi mobile base, recorded in Jan. 2026 as the PoC's 41 s evaluation video: the base's arm picks the parts, the base carries them to the assembly table, two fixed SO-101 arms assemble them, and an inspection camera reports pass/fail.

- **Design** — Code-as-Policies: the task sequence is fixed; Gemini, a vision-language model, fills a code skeleton in the prompt with each skill's code offline (over the robot's perception and control primitives) for a skill library; at run time it reads the camera frame and selects the skill, whose code then drives the arms on LeRobot, and the base moves by line following.

Agentic AI — LLM Agents for System Operation (Robotics & V2G)

[Skill VAE, Diffusion Decoder, In-Painting] LLM-Guided Skill Diffusion System for Test-Time Policy Adaptation (LDuS) 2023 – 2024

- **Achievements** — NeurIPS 2024 Main Conference Track, **second author**
- **Role: Leader** — problem definition, algorithm design, architecture, experiment design; robot-task validation on Meta-World and multi-stage Meta-World (3 long-horizon goals).
- **Built** — LDuS steers a pretrained skill diffusion policy at test time with loss-guidance code that an LLM writes from a natural-language context (speed, energy, or spatial constraints).
- **Model** — A hierarchical skill model pretrained on goal-annotated expert trajectories: a VAE skill encoder maps an h-step state–action segment to a skill embedding, a skill prior predicts it from the state and the language goal encoded by CLIP (contrastive language–image pretraining), and a skill diffusion planner (a U-Net denoiser as in Diffuser) reconstructs the segment from the embedding. At test time GPT-3.5 writes the loss code from the context, the loss gradient guides every denoising step, and GPT-4 judges contact from the camera frame to gate constraints (such as a force constraint) that apply only while touching the object.
- **Design** — Sequential in-painting conditions target trajectories on past state–action sequences, with no policy retraining. A closed loop validates the generated code and the guided trajectories against the context, regenerating faulty code and raising the guidance frequency until the trajectories satisfy the context.
- **Result** — On Meta-World speed constraints, LDuS reaches a 94.6% success rate and 87.4 context reward, against 76.4% and 69.8 for Diffuser+Guidance (the same gradient guidance with a hand-written loss). On multi-stage Meta-World tasks, it reaches 84.0% and 82.0, against 36.7% and 45.9.

[LLM Agents, Drift Detection, RL Scheduler] Agentic Scheduling Framework for V2G Fleet Operation (AgentV2G) 2026 – present

- **Achievements** — Hyundai Motor · Kia V2X project · interim report delivered Oct. 2026
- **Role: Leader** — (full cover) problem definition, algorithm design, implementation, experiments; construction of the V2G fleet-operation environment (Jeju two-settlement pilot-market rules).
- **Built** — V2G-OpSim, the charging-hub simulator built on the Jeju day-ahead and real-time two-settlement pilot rules; AgentV2G is designed to run the hub's electricity-market bidding and charge/discharge dispatch on it.
- **Design** — An LLM agent detects drift on two axes, electricity price and vehicle use. Drift is declared when observations that differ from the reference accumulate past a set amount. On the price axis the agent selects a tool that corrects the price forecast; on the vehicle axis it selects an RL scheduler trained for the new condition; when no fitting tool exists, it creates one from the accumulated observations.
- **RL scheduler** — The condition-specific scheduler is a SAC (soft actor-critic) policy for the real-time dispatch stage. A shared MLP embeds each EV's observation, and the mean of these local embeddings forms the fleet embedding. A shared policy head reads the local and fleet embeddings to emit each EV's charge or discharge ratio, so the policy is independent of the fleet size. The action is learned as a residual over rule-based dispatch and can be projected onto the allowed band; the Jeju pilot-market settlement penalties (band deviation, unmet charging) and battery degradation cost enter the actor loss as a regularization term.
- **Project result** — In the project's interim report (Oct. 2026), a separate fleet-common policy trained on 1,000 EVs ran on 10,000 EVs without retraining at 66 ms (P90) per dispatch step; the project's target is 90 s.

Systems Engineering — Simulators, Testbeds, Protocols & Deployment

Federated Learning & Edge Deployment

[Federated Learning, Knowledge Distillation, YOLOv8n] Edge–Server Collaborative Inference Pipeline for Real-Time Traffic Sign Recognition 2025

- **Achievements** — **Delivered to** Acryl Inc.
- **Role: Leader** — problem definition, algorithm design, implementation, experiments, testbed.
- **Built** — Two-tier road-sign recognition for road-traffic monitoring, flagging damaged or occluded signs per frame. The edge YOLOv8n was produced by cooperative federated learning: FedAvg over the devices' daytime traffic-sign data, then server-side knowledge distillation from a YOLOv8l pretrained on night traffic-sign data through a distillation layer on the backbone's P5-level features with a dimension-matching projection.

- **Model** — YOLOv8n on the edge and YOLOv8l on the server. The Flower-based federated-learning framework was extended on the client side with federated RL and per-client non-IID (non-independent, non-identically distributed) data composition; Flower's per-epoch learning-rate schedule was made adjustable per round and set to cosine annealing, and image-transform augmentation was added; the schedule and augmentation raise CIFAR-10 accuracy from 82% to 84% (86.1% with STL-10 auxiliary data) on the framework's supervised benchmark.
- **Design** — The distilled YOLOv8n on the FPGA-based NPU processes routine frames in real time. When a frame's confidence falls below 0.3 (e.g., at night or in rain), the next frame is forwarded to the larger server-side model for federated inference, improving accuracy on the hard frames without adding edge latency to the easy ones. Night driving data is assumed to be scarce on the devices, so night is treated as the unseen condition: night images were synthesized from the daytime set with OpenCV in easy and hard modes, and the night-pretrained large model supplies the knowledge the federated model lacks, without moving device data.
- **Result** — Night-image accuracy is 80.1% (easy) and 36.6% (hard), 20 and 16 points over federated learning alone.

Simulators, Testbeds & Surrogates

[C++ Packet Simulation, NS-3, 802.11ac] Network Environments of Increasing Fidelity for Rate-Control Policy Pretraining and Validation (S-Sim, NS-3, NiL Testbed) 2020 – 2026

- **Role: Leader** — implementation, experiments, testbed.
- **Built** — XferQoE's training backend and validation path: a custom µs-resolution packet-level C++ simulator with no protocol stack (S-Sim), a full-stack NS-3 environment, and a Raspberry Pi network-in-the-loop (NiL) testbed (2026) that puts a real wireless link in the control loop. On the testbed, Raspberry Pi 4 senders share one IEEE 802.11ac link through an access point to an edge server (one sender for robot control, up to 16 for monitoring).
- **Design** — The three environments share one state-input, rate-output interface and replay the same bandwidth trace (Puffer), so a policy pretrained entirely in simulation runs on all three unchanged; S-Sim adds parallel multi-environment rollout and replay of eight bandwidth traces (four synthetic, four real-world). On the testbed, C/C++ TCP/UDP daemons implement episode-level clock synchronization with drift measurement, sender-side virtual qdisc queue modeling, and frame-deadline expiry accounting.
- **Result** — On the testbed, policy inference and rate computation take 6.7 ms of each 1 s control interval.

[NS-3, FIO] NVMe-over-TCP Disaggregated Storage Simulator with a Dual-Output DNN Surrogate 2019 – 2021

- **Achievements — Delivered to** Samsung Electronics · US 12,210,760 B2 · US 12,650,769 B2 · SW Reg. C-2020-037999
- **Role: Leader** — problem definition, algorithm design, experiments.
- **Built** — The simulator reproduces a disaggregated NVMe-over-TCP cluster in software, so storage management policies can be tuned for the mixed device generations and configurations a production data center runs.
- **Model** — The surrogate takes a 15-bin histogram of the objects' required IOPS (1,000-IOPS bins) and predicts serviced IOPS and latency with one DNN, trained with an L-infinity loss on serviced IOPS and an L1 loss on latency over FIO logs in which each object's required IOPS is set at random.
- **Design** — FIO measures per-node serviced IOPS and latency against required IOPS, and one DNN surrogate spans device specifications. Devices are organized under a platform that aggregates them in a two-level hierarchy, so the environment is reconfigured from a system specification rather than rebuilt. A tier-level object-migration policy is evaluated across homogeneous and heterogeneous compositions of the measured devices under FIO workloads.
- **Result** — Predicted IOPS and latency responses are consistent with FIO measurements on the four profiled NVMe-over-TCP configurations, from an RC500 SSD on 10 GbE to a Samsung PM1725a SSD on a 40 GbE NIC.

[TCP/UDP, Raw-Socket Sniffer, 802.11ac] Low-Latency Wireless Streaming Testbed for the Samsung QLED TV One Connect System 2018

- **Achievements — Delivered to** Samsung Electronics
- **Role: Leader** — implementation, experiments, testbed.
- **Built** — The testbed reproduces the Samsung One Connect link that feeds a QLED TV; transmitter and receiver both run on Linux hosts with Broadcom BCM4360 wireless NICs.
- **Design** — Layer-by-layer delay analysis of the standard stack for 4K 60 Hz video. Three C/C++ components generate and measure the traffic: a TCP sender–receiver with buffer-based flow control, a UDP sender–receiver with bitrate-paced transmission, and a raw-socket sniffer measuring RTT, throughput, packet loss, duplication, and reordering.
- **Result** — Bitrate-paced UDP transmission sustains 4K 60 Hz frames at 6 ms average RTT (TCP path: 11 ms).

Evaluation Environments & Training Framework

[Meta-World, ManiSkill3, YOLO Scoring] Multi-Application QoE Evaluation Environments for Network Policy Learning 2024 – 2026

- **Role: Leader** — implementation, experiments.
- **Built** — Application layer of XferQoE: three downstream environments that consume a network-delivered stream and score it with a quality model of their own. For the RFM tasks, SmolVLA was fine-tuned on 250 MIKASA-Robo-VLA Intercept demonstrations for 20,000 steps, with only its action expert and state input layer trained.
- **Design** — Where a learned model decides quality, delivery loss and staleness propagate through that model's decision boundary rather than through a designer's formula. Robot control (Meta-World, ManiSkill3, and robot foundation models on LIBERO and MIKASA), networked monitoring (16 devices scored live by a YOLO detector), and video streaming (live and VoD) each weigh the QoS factors differently.

[SAC, JIT/vmap] Unified JAX/Flax Training Framework for Network Control Policies 2020 – 2026

- **Role: Leader** — implementation.
- **Built** — A shared soft actor-critic (SAC) implementation supports all nine learning-based methods compared with XferQoE and three hand-crafted controllers, so every method is compared under identical conditions.
- **Design** — Encoders, decoders, and actor-critic are organized under one three-tier abstraction (layer / block / module) with a single dataclass I/O convention, so supervised learning and RL share components. Fixing hyperparameters as static arguments compiles the training step into one JIT kernel, and per-task parameter stacks vectorized with vmap process every parallel environment in a single forward pass.

[Gym Environment (EV2Gym), Market-Rule Presets] V2G Fleet-Operation Evaluation Environment for Two-Settlement Markets (V2G-OpSim) 2026 – present

- **Role: Leader** — problem definition, implementation.
- **Built** — Charging-hub environment implementing the Jeju two-settlement pilot rules, with EU and NEM (Australian National Electricity Market) presets.
- **Design** — Korea Power Exchange (KPX) SMP series and EV charging-session records drive it. Price, session behavior, and the rules themselves drift alone or together, so a scheduler is scored on episode-level operating quality and on how fast it recovers after the drift on the Jeju pilot-market rules.

Delivered Services & Tools

[PostGIS, HTTP API, Flutter] Database-Backed Network Selection Application for Mobile Terminals 2024

- **Achievements — Delivered to** Polaris AI
- **Role: Leader** — algorithm design, implementation.
- **Built** — The application recommends the network a public-safety LTE (PS-LTE) terminal should use, PS-LTE or a commercial LTE band, in areas where the disaster-response network and the railway network share a band and interfere.
- **Design** — A PostGIS database of over 200,000 normalized field measurements backs the ranking of 6 LTE bands (10 carriers) by RSRP, RSRQ, and SINR, using the nearest measurements in sparsely covered areas. A server provides that ranking and the threshold-based handover decision between PS-LTE and the commercial network via an HTTP API, and a Flutter client presents the recommendation on the terminal.

[MATLAB, Monte Carlo] GUI-Based Analysis Tool for Direct GPS P(Y) Code Acquisition under Wideband Jamming 2013

- **Achievements — Delivered to** ADD
- **Role: Leader** — implementation, experiments.
- **Built** — MATLAB GUI tool that simulates direct acquisition of the military GPS P(Y) code, bypassing the C/A code; Monte Carlo analysis of mean acquisition time across jamming-to-signal ratios of 0–42 dB gives the maximum jamming level under which acquisition still succeeds.

Protocol & Packet Design

[MQTT, Packet Design] IoT Network Protocol and Packet Structure for an Intelligent Gateway 2014 – 2017

- **Achievements** — US 10,742,742 B2, held jointly with Samsung Electronics (**third of four inventors**)
- **Role: Leader** — algorithm design (protocol specification and packet-structure redesign).
- **Built** — The gateway–device network protocol was defined and specified end to end: an MQTT-based message protocol whose packet structure was redesigned around fixed-size sub-resource identifiers.

- **Design** — A controlling apparatus assigns each device one fixed-size identifier item per level of its URI hierarchy; the device stores it, requests and commands carry it, and both ends keep the mapping in a resource access table.
- **Result** — Packets carry the fixed-size identifier instead of the long, variable-length URI that wastes network capacity, so packet size decreases and transfer efficiency improves.

Deep Learning Models

[GCN Encoder, Mapping Decoder, REINFORCE] RL-Based Graph Embedding and Mapping System for Virtual Networks (Gemma) 2020 – 2021

- **Achievements** — IEEE Access (2021), **second author** · KR 10-2629731 · US 12,047,250 B2 (**first inventor**)
- **Role: Participant** — algorithm design (the network structure; the mapping method, patented with the applicant as first inventor), experiment design.
- **Built** — Gemma maps a requested virtual network onto the physical substrate that will host it; this is the resource-allocation problem behind data-center tenancy, ISP network slicing, and service-function chaining.
- **Model** — A GCN (graph convolutional network) encoder embeds the virtual and the substrate networks. A node-mapping decoder selects the top-ranked virtual node and scores it on each substrate node through two MLP rank functions (two-stage mapping); the encoder and decoder are trained end to end by a modified REINFORCE that maximizes the problem-specific expected reward by gradient ascent.
- **Design** — One learned structure covers both request types without being rebuilt: static (offline batch) requests that hold their allocation once placed, and dynamic (online) ones that arrive over time and are judged on placement speed. Iterative decoder ranking lets the same model handle substrates of varied structure and capacity.
- **Result** — Gemma outperforms heuristic and learning-based methods on data-center online arrivals and ISP offline batch slicing, and re-maps allocations under substrate capacity changes at a tunable reconfiguration cost.

[LSTM, Pretrained Embedding Transfer] Activity-Stream Embedding Model for Continual Bug-Fix Time Prediction (DASENet) 2019 – 2020

- **Achievements** — IEEE Access (2020), **first author**
- **Role: Leader** — problem definition, algorithm design, implementation, experiments.
- **Built** — DASENet predicts how long an open bug will take to fix and revises the estimate as developer activity arrives, so release planning and staffing can be adjusted while the bug is still open.
- **Model** — Each bug report becomes a sequence of per-day activity bins. Per bin, deep BLSTMs (bidirectional LSTMs) encode the user comments and the system records, an MLP encodes the metadata, and a merging MLP yields the bin embedding; a residual LSTM (RLSTM) sequence network over the bin embeddings yields the activity-stream embedding; the whole network is pretrained through an MLP and softmax over bug-fix-time classes. A task MLP stacked on the pretrained network is then trained separately for each task, without end-to-end fine-tuning.
- **Result** — Trained on about 900 samples (2% of the full dataset) on top of the pretrained activity-stream embedding, a variant task model stays competitive with full-data models, showing the learned structure transfers.

PUBLICATIONS

- The five works below are listed in full; 35 further works from 2012 to 2018 (3 other international journals, 5 domestic journals, 18 international conferences, 9 domestic conferences) are on Google Scholar.
- Since 2018, the research has focused on learning-based network control and policy transfer with Prof. Honguk Woo; earlier work was conducted in two other groups at the same university — content-centric networking, IoT, and vehicular communication with Prof. Ikjun Yeom, and GNSS signal synchronization with Prof. Seokho Yoon.

Google Scholar: scholar.google.com/citations?user=GhwtxtgAAAAJ

Under Review

1. **Youngseok Lee** and Honguk Woo, "XferQoE: QoS-to-QoE transfer for adaptive rate control," submitted to *IEEE Internet of Things Journal*.

Top-Tier Conferences

1. Woo Kyung Kim, **Youngseok Lee**, Jooyoung Kim, and Honguk Woo, "LLM-based skill diffusion for zero-shot policy adaptation," in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, Main Conference Track.

International Journals (SCIE)

1. **Youngseok Lee**, Woo Kyung Kim, Sung Hyun Choi, Ikjun Yeom, and Honguk Woo, "Repot: Transferable reinforcement learning for quality-centric networked monitoring in various environments," *IEEE Access*, vol. 9, pp. 147280–147294, 2021.
2. Minjae Park, **Youngseok Lee**, Ikjun Yeom, and Honguk Woo, "Gemma: Reinforcement learning-based graph embedding and mapping for virtual network applications," *IEEE Access*, vol. 9, pp. 105463–105476, 2021.
3. **Youngseok Lee**, Suin Lee, Chan-Gun Lee, Ikjun Yeom, and Honguk Woo, "Continual prediction of bug-fix time using deep learning-based activity stream embedding," *IEEE Access*, vol. 8, pp. 10503–10515, 2020. **(33 citations)**

PATENTS

Granted: 8 (United States 5, Republic of Korea 3) · First inventor on 4

- The eight granted patents cover six distinct inventions.

Invention	Registration(s)	Granted	Notes
Graph Embedding-Based Virtual Network Mapping Method	KR 10-2629731 · US 12,047,250 B2	KR 2024 · US 2024	First inventor · SKKU · Related: Gemma, IEEE Access (2021)
Network Monitoring System	KR 10-2574070 · US 12,556,492 B2	KR 2023 · US 2026	First inventor · SKKU · Related: Repot, IEEE Access (2021)
Object Storage System, Migration Control Device, and Migration Control Method	US 12,210,760 B2	2025	Held with Samsung Electronics · fourth of four inventors
Simulation Method and Modeling Method	US 12,650,769 B2	2026	Held with Samsung Electronics · fourth of four inventors
Electronic Apparatus, System for Internet-of-Things Environment and Control Method Thereof	US 10,742,742 B2	2020	Held with Samsung Electronics · third of four inventors
Simulation Apparatus and Method for Time Synchronization in Spread Spectrum System	KR 10-1305307	2013	SKKU · GNSS signal-synchronization work · third of three inventors

FUNDED PROJECTS

- **20 funded projects** — Principal investigator 1 · technical leader 10 · researcher 9; sponsors span MSIT/MSIP, MOE, Samsung Electronics, Hyundai Motor · Kia, Polaris AI, and ADD

MSIT = Ministry of Science and ICT; MSIP = its predecessor, the Ministry of Science, ICT and Future Planning (before 2017); MOE = Ministry of Education; NIPA = National IT Industry Promotion Agency; IITP = Institute for Information & Communications Technology Planning & Evaluation. Roles: principal investigator (연구책임자), technical leader (실무책임자), researcher (참여연구원).

Reinforcement Learning — Resource Management & Remote Model Serving

Project	Sponsor / Program	Role	Period
이기종 협업-피지컬AI SDF 특화 기반 모델 연구개발	MSIT / NIPA, Collaborative-Intelligence Physical-AI SW-Platform R&D Ecosystem — Hosted by SKKU	Researcher	Aug. 2026 – Dec. 2030 (ongoing)
실세계 범용 체화 지능 실현을 위한 반응형 스킬 디퓨전 연구	MSIT / Core Research (Leap)	Researcher	Mar. 2026 – Feb. 2029 (ongoing)
피지컬AI 핵심기술 PoC — 이기종 협업-피지컬 AI SDF 특화 기반 모델 연구개발 (제2과제)	MSIT / NIPA, Industry-Specific Physical-AI Leading-Model PoC	Researcher	Sep. 2025 – Dec. 2025
지속학습을 통한 인사이트 고속 추론 연구	MSIT / IITP, Human-Centered Next-Gen AI	Researcher	Apr. 2025 – Dec. 2029 (ongoing)
멀티모달 스킬 전이 기반 정책 일반화 기법 연구	MSIT / Young Researcher (Excellent)	Researcher	Mar. 2023 – Feb. 2026
동적인 디바이스 환경에서 적응적 연합학습기술 개발	MSIT / IITP, Next-Gen AI Core	Technical leader	Apr. 2021 – Dec. 2022

Agentic AI — LLM Agents for System Operation (Robotics & V2G)

Project	Sponsor / Program	Role	Period
V2X 운영 최적화를 위한 강화학습 기반 기술 개발	Hyundai Motor · Kia	Technical leader	Mar. 2026 – Mar. 2027 (ongoing)
개성 형성이 가능한 에이전트 플랫폼 기술 개발	MSIT / IITP, Human-Centered AI Core	Researcher	Apr. 2022 – Dec. 2026 (ongoing)
오픈 도메인 멀티모달 자기주도 인공지능 기술 개발	MSIT / IITP, Human-Centered AI Core	Researcher	Apr. 2022 – Dec. 2026 (ongoing)

Systems Engineering — Simulators, Testbeds, Protocols & Deployment

Project	Sponsor / Program	Role	Period
차세대 컴퓨팅환경을 위한 스토리지시스템 및 응용프레임워크 최적화 기술 연구	Samsung Electronics	Technical leader	Aug. 2019 – Jul. 2022
재밍대응 식별에 의한 위성항법신호 복원기법 연구	Agency for Defense Development (SN-42)	Technical leader	Mar. 2013 – Oct. 2013
재난안전통신망과 철도망 전파간섭 지역에서 AI기반 대규모 PS-LTE 단말기 접속방법 개선에 관한 연구	Polaris AI	Technical leader	Aug. 2024 – Dec. 2024
데이터센터-엣지 NPU 간 연합 추론/학습 기반 대규모 인공지능 응용/개발을 용이하게 하는 SW 프레임워크 개발	MSIT / IITP, SW Computing Industry	Technical leader	Apr. 2021 – Jun. 2025
저 지연 무선 네트워크 프로토콜 개발	Samsung Electronics	Technical leader	Sep. 2017 – Sep. 2018

Early Work — Networking, GNSS, IoT & Media Platforms

Project	Sponsor / Program	Role	Period
소프트웨어 정의 커넥티드 카	MSIP / Individual Research (Strategic)	Technical leader	Dec. 2016 – Nov. 2019
사물인터넷을 위한 지능형 게이트웨이	MSIP / Global Research Program	Technical leader	Nov. 2014 – Jun. 2017
유무선 통합망을 위한 콘텐츠 중심 네트워크 기술	MSIP / Basic Research Program	Technical leader	Mar. 2014 – Apr. 2015
스마트 TV 2.0 소프트웨어 플랫폼	MSIP / ICT & Broadcasting R&D	Researcher	Mar. 2014 – Aug. 2016
BOC 신호 추적을 위한 새로운 상관함수에 대한 연구	MOE / Global Ph.D. Fellowship	Principal investigator	Mar. 2013 – Feb. 2018
초정밀 위치 기반 서비스를 위한 여러 대역 신호 및 이종 항법위성 기반 신호처리기술	MSIP / Core Individual Research	Researcher	Mar. 2013 – Oct. 2013

SOFTWARE COPYRIGHT REGISTRATIONS

Each program below is registered with the Korea Copyright Commission (registrant: Research & Business Foundation, Sungkyunkwan University); the numbers cited in Development Experience refer to this list.

Reg. No.	Title	Created	Registered
C-2023-048806	Hierarchical Meta-Reinforcement Learning Model	Oct. 19, 2023	Nov. 3, 2023
C-2021-054170	Federated Learning Performance Evaluation for Heterogeneous Datasets (Flower Framework)	Nov. 24, 2021	Dec. 9, 2021
C-2021-041557	Behaviour Analysis Program for Risk-Sensitive Reinforcement Learning	Sep. 16, 2021	Oct. 20, 2021
C-2021-035993	Replay-Buffer Sharing for Accelerated Personalized Federated Reinforcement Learning	Aug. 25, 2021	Sep. 8, 2021
C-2020-039571	SUMO Simulator-Based Clustering Framework	Oct. 15, 2020	Nov. 4, 2020
C-2020-038567	Backscatter Network Simulator for the EPC Gen2 Protocol	Oct. 22, 2020	Oct. 29, 2020
C-2020-037999	ns-3-Based NVMe-oF Disaggregated Storage Simulator for Reinforcement Learning	Sep. 5, 2020	Oct. 27, 2020
C-2016-027920	Signal Synchronization Performance Measurement for Galileo under Noisy Multipath	Oct. 28, 2016	Nov. 18, 2016
C-2016-020663	Cache Performance Measurement Program for Content-Centric Networking	Jul. 14, 2016	Aug. 24, 2016

HONORS & FELLOWSHIPS

Year	Honor / Fellowship	Awarded by
2014 · 2015	Outstanding Research Achievement, Global Ph.D. Fellowship Program	Ministry of Education / NRF Korea
Mar. 2013 – Feb. 2018	Global Ph.D. Fellowship — Full 5-year term; 200 selected nationwide in 2013	Ministry of Education / NRF Korea
Dec. 2012	IEEE Seoul Section Student Paper Contest — Bronze Award	IEEE Seoul Section
Nov. 2012	Honorable Mention, 9th Undergraduate Thesis Presentation Contest	Sungkyunkwan University
Mar. 2008 – Feb. 2013	National Science & Engineering Scholarship — Full tuition, 8 semesters	MEST / Korea Science & Engineering Foundation